ZUBIN JAIN

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PROFESSIONAL SUMMARY

ASIC Engineer with hands-on experience across RTL design, UVM verification frameworks, and microarchitecture development for high-performance NPU architectures. Proven ability to thrive in fast-paced startup environments, accelerate FPGA deployment pipelines, and leverage cutting-edge LLM configurations to build robust Agentic hardware automation workflows.

TECHNICAL SKILLS

Languages: SystemVerilog, Verilog, Python, C/C++, Assembly, TCL, Perl, LabVIEW

Methodologies: RTL Design, UVM, Test Planning, Coverage Closure, HW/SW Co-Design

Protocols & Arch: AXI, APB, PCIe, Systolic Arrays, NPU, STM32 MCU Microarchitecture

Tools & EDA: Vivado, Cadence EDA Tools, Power BI, Jenkins, Git, ModelSim

AI & LLM Tools: Claude (Anthropic) — fluent in agentic RTL workflows, prompt engineering, and automation; Codex — fluent in AI-assisted code generation, legacy refactoring, and hardware toolchain scripting

WORK EXPERIENCE

AI ASIC Engineer — Expedera *May 2024 – Present*

- Design and implement high-throughput SystemVerilog RTL modules for specialized NPU architectures, integrating AXI, APB, and PCIe communication protocols.
- Pioneered the integration of Claude and Codex into Agentic RTL workflows, automating legacy code refactoring and accelerating early-stage microarchitectural exploration.
- Lead comprehensive verification efforts using UVM methodology, establishing test plans, managing model generation, and driving full functional coverage closure.
- Conduct static timing analysis and microarchitectural optimizations to resolve critical paths in high-performance hardware blocks.
- Architected and automated robust continuous integration/regression infrastructure utilizing Jenkins, Python, and Perl.
- Streamlined hardware-in-the-loop validation by automating FPGA synthesis and deployment workflows through Vivado and custom TCL scripting.

Firmware & PCB Engineer (Part-Time) — Tribecar *Jan 2024 – May 2024*

- Served as the sole engineer executing firmware development and electronic hardware redesign for commercial vehicle tracking systems.
- Overhauled PCB layouts to handle high-current operations safely and mitigate electromagnetic interference to secure signal integrity.
- Reverse-engineered and debugged low-level STM32 microcontroller firmware to locate and rectify legacy design flaws.
- Resolved a critical firmware race condition bug, establishing 100% reliability for the fleet's remote vehicle unlock functionality.

NPI Intern — AMD *Jan 2023 – Aug 2023*

- Developed data processing utilities to clean and parse mass Automated Test Equipment (ATE) raw silicon validation datasets.
- Engineered Python data pipelines that automated manual Excel workflows, cutting analysis cycle times and accelerating engineering reporting.
- Constructed interactive Power BI analytics dashboards to extract and visualize Yield and product classification insights.
- Gained end-to-end operational exposure to structural semiconductor testing, test program flows, and silicon characterization processes.

EDUCATION

National University of Singapore (NUS) *Aug 2020 – May 2024*

Bachelor of Electrical Engineering (Honours with Distinction) | GPA: 4.1 / 5.0

RESEARCH & ENGINEERING PROJECTS

Hardware Acceleration of SLAM Algorithms — *Bachelor's Thesis* *Academic Year 2023 – 2024*

- Designed and evaluated FPGA-based acceleration architectures for spatial mapping and localization frameworks at the edge.
- Co-designed and profiled hardware implementations of Fast-SLAM, Graph-SLAM, and Kalman filter blocks.
- Successfully deployed the accelerated system architecture onto an AMD Xilinx Kria KV260 SoC FPGA paired with ESP32 sensor modules.

Open-Source Contributor — *Verilator* *2024 – Present*

- Active contributor to Verilator, the industry-leading open-source SystemVerilog simulator and lint tool.
- Leveraged Codex to identify, diagnose, and fix compilation-blocking bugs in the Verilator codebase, resolving issues that were preventing the project from successfully building across targeted toolchain configurations.
- Contributed patches upstream, engaging with the maintainer community through code review and issue tracking workflows.

Project Mynah — *Singapore Propulsion Lab* *2022 – 2023*

- Engineered the core avionics and telemetry flight computer systems for a high-altitude sounding rocket deployment.
- Acquired hands-on regulatory compliance experience by spearheading the engineering validation required to obtain FCC radio certification.
- Supported real-time ground station telemetry logistics and launch operations successfully at FAR 2023.

Research Intern — *Green IC Lab, NUS* *Jan 2023 – Aug 2023*

- Characterized ultra-low power mixed-signal ADC/DAC silicon performance chips under the guidance of Prof. Massimo Alioto.
- Built automated bench measurement and data acquisition configurations using LabVIEW software to interact with laboratory test instruments.
- Worked with industry-standard Cadence tools to execute basic IC physical layout verification tasks.

Health Study Data System — *NUS School of Nursing* *Jan 2022 – Jul 2022*

- Built a video-based motion-tracking system optimized for remote clinical health study data collection.
- Implemented custom joint-tracking and skeletal analysis algorithms using computer vision principles to evaluate physical mobility issues.

ADDITIONAL INFORMATION

Leadership: Founder & Editor-in-Chief of The Sengkang Sci-Fi Journal — established the publication from the ground up, assembled and managed a team of writers, editors, and graphic designers, oversaw full editorial pipelines and publication curation, and led donor outreach and sponsorship courting to sustain the journal’s operations.

Work Authorization: Eligible to work immediately in Singapore (Citizen) and India (OCI cardholder) without visa constraints.

Collaboration: Deep experience working across international time zones within distributed engineering and startup environments.